# PROJECT AND TEAM INFORMATION

## Project Title

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| *HealthQueue - Smart Patient Management System* |

## Student / Team Information

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| *Team Name:*<br>*TechNova*<br>*Team #* | *Avengers* |
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# PROPOSAL DESCRIPTION (10 pts)

## Motivation (1 pt)

*Every day hospitals deal with a lot of patients. If the handling of patient data is done manually, this can consume time and lead to errors. Urgency cases also have to be given prompt help. In our project we aim at tackling this issue by developing a simple Computer Patient Management System in C++. The system will help in registering the patients, searching the information about patients, and maintaining the waiting list. Besides, the system will give priority to the urgent cases. The simple data structures like Queue, Priority Queue, and Linked List will be used in this project. These data structures will teach us how to deal with patients easier. The importance of the project is that it connects our theoretical knowledge of Data Structures obtained at the classroom with the real life problem. In addition, this important project will improve our programming, problem-solving and team-working skills.*

## State of the Art / Current solution (1 pts)

*Some hospitals still have systems that require the use of either manual registers or computerized systems to keep track of the patients' information. While manual registers are easier to use, the process of searching and updating patients' information is time-consuming and there's a possibility of making errors or losing information.*

*Some hospitals are using the advanced software applications for patient registration and appointments, as well as health records. But, such software may be quite technical in use and require many resources to operate.*

*The goal of this project is to develop a simplified solution for educational purposes. It is going to cover the basic aspects of managing patients' information using C++ and data structures. The system will provide the following services: registration of patients, their search and waiting-list management and file management features. It will not be complex enough, so it can be used in small hospitals and clinics.*

## Project Goals and Milestones (2 pts)

*The main aim of the project is to create an uncomplicated yet effective Hospital Patient Management System through C++ and Data Structures. The software will assist in keeping a record of patients' details and thus make the treatment process much more streamlined.*

*>Project Goals:*

1. *To be able to register patients and save their particulars*
2. *To be able to search and find patients' details easily*
3. *To be able to manage normal patients with the help of Queue*
4. *To be able to manage emergency patients with the help of Priority Queue*
5. *To be able to maintain the records of patients with the help of Linked List*
6. *To be able to store complaints from patients with the help of file handling*

*>Project Milestones:*

*1. Select the topic of the project and understand the problem*

*2. Collect the data and come up with the features of the system*

*3. Design the basic menu and mission statement of the program*

*4. Implement patient registration and patient search*

*5. Implement Queue for the waiting list*

*6. Implement Priority Queue for emergency patients*

*7. Implement Linked List for maintaining patients' records*

*8. Add file handling for patient complaints*

*9. Test all the functionalities and remove bugs*

## Project Approach (3 pts)

*The program is going to be made using C++ programming language as well as simple datatypes for the data structures. The application will have a menu and will make it easier for a hospital staff member to handle the patients' information.*

*Data such as patient ID, name, age, disease, doctor's name, and patient's contacts will be kept and handled using data structures.*

*A linked list will be used to keep patient data, whereas a queue will be used to hold normal patients. Various types of priority queues will be used to handle emergency patients in accordance with their priority class.*

*Searching and sorting algorithms will be applied to find a records and arranging them as well.*

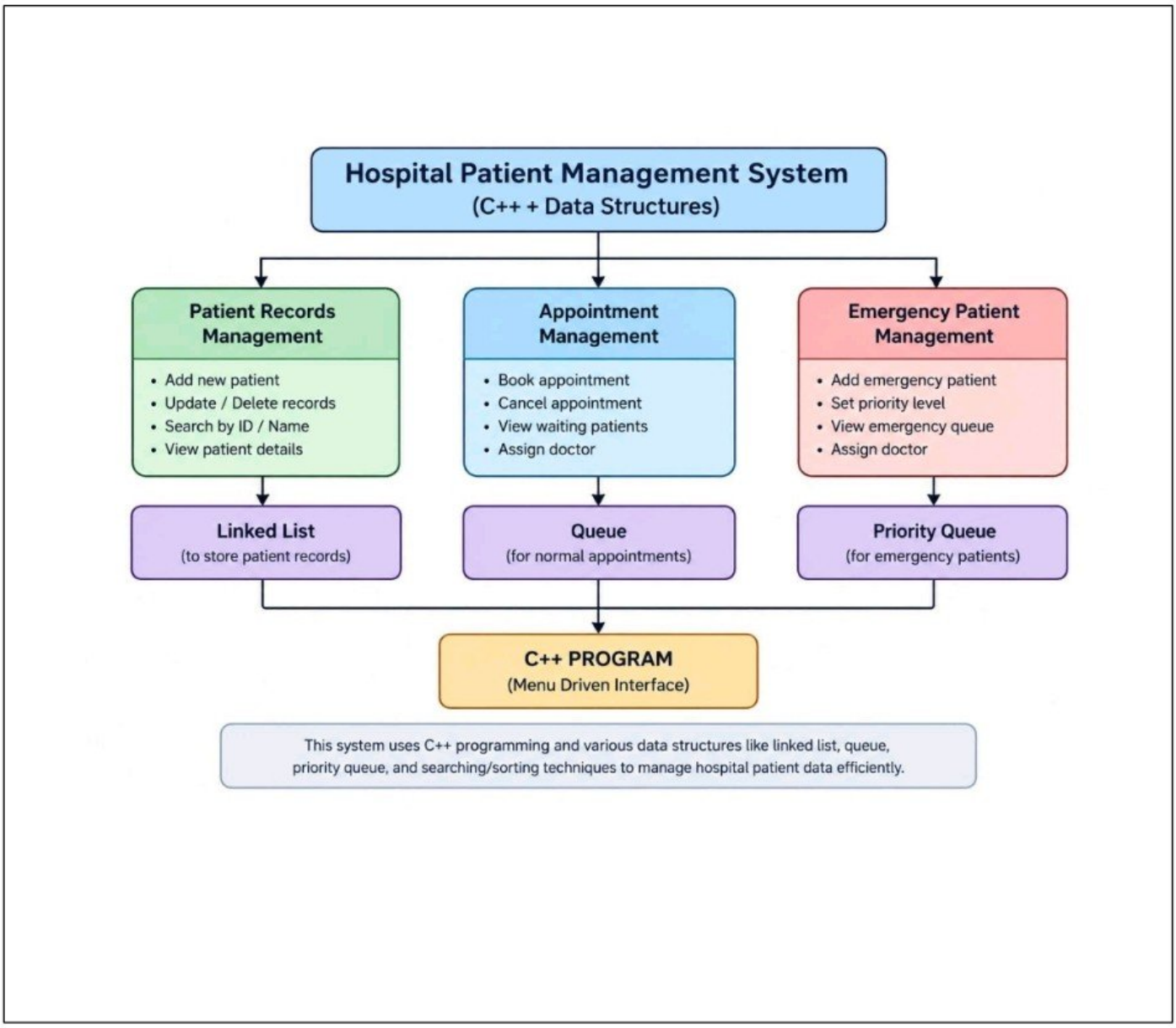
*The design and implementation will be carried out using such C++ compilers as Visual Studio Code and Code::Blocks.*

## System Architecture (High Level Diagram)(2 pts)

*The system is developed using C++ and Data Structures.*

*Patient records are managed using a Linked List, normal appointments are handled using a Queue, and emergency patients are managed using a Priority Queue.*

*These modules are connected through a simple menu-driven C++ program to manage patient information efficiently.*




## Project Outcome / Deliverables (1 pts)

*The last project creates a simple Hospital Patient Management System which is easy to use. The users will be able to add, change, remove, find and show the details of patient records. Besides, the system will be responsible for normal appointments which are organized by means of queue and for emergency patients via priority queue.*

*The basic results of the project will be operational programme C, adapting different data structures, having process related to patient and appointment management, as well as searching and sorting actions. The project will show how C and Data Structures can be applied practically to solve a hospital related issue.*

## Assumptions

*The application was devised for either a small or medium hospital. It is assumed that the patient details inserted by the operator are correct. Every patient will be issued a unique identification number. The application will provide basic management functions only as it will not have online payments, real-time doctor's availability, and online registration. The data concerning patients will be managed during the program execution using C++ data structures. Emergency patients will be given more attention than regular patients. The system is meant primarily for educational purposes to exemplify C++ and Data Structures usage in the real-world scenario.*

## References

*1.C++ fundamental principles and library manuals.*

*2.Data Structures and Algorithms – fundamental concepts of Linked List, Queue, Priority Queue, Search and Sort methods.*

*3.Learning material and literature on C++.*

*4.Class material and notes shared by faculty members.*